Name: ____________________________

Date: ____________________________

Mr. Rawson • Y9 MYP Physics

1.10 – Natural Frequency and Resonance

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define **natural frequency** as the frequency at which an object oscillates freely when disturbed.
- b) I can distinguish between **natural frequency** and **driving frequency**.
- c) I can explain how **resonance** occurs and why it leads to large amplitudes.
- d) I can predict how changing mass or stiffness affects natural frequency.

Theory

Every physical object has a **natural frequency**, which is the specific rate at which it prefers to vibrate when it is set in motion and then left alone. Think of a pendulum: if you pull it back and let go, it swings back and forth at its own steady pace. This pace depends entirely on the object's physical properties, such as its length, mass, or stiffness. If you were to change the length of the pendulum, its natural frequency would change because the physics of the system has changed.

When an external force pushes or pulls on an object repeatedly, that push happens at a specific rate called the **driving frequency**. For example, if you are pushing a child on a swing, the number of times per second you give the swing a push is the driving frequency. You can choose to push slowly or quickly, regardless of how fast the swing wants to go naturally.

Resonance occurs when the driving frequency exactly matches the natural frequency of the object. When these two frequencies align, energy is transferred into the system most efficiently. Each push adds a little more energy to the motion at the perfect moment in the cycle, causing the **amplitude** (the size of the oscillation) to grow larger and larger with every cycle. This is why you can make a swing go very high just by timing your pushes correctly.

This effect can be powerful enough to cause damage. If the amplitude becomes too large, the forces within the material may exceed its strength, leading to structural failure. A famous example is the collapse of the Tacoma Narrows Bridge in 1940. Wind gusts acted as a driving force that matched the bridge's natural frequency, causing it to twist and vibrate with increasing amplitude until it broke apart. Similarly, a singer can shatter a wine glass by holding a note that matches the glass's natural frequency.

The natural frequency of an object is determined by its mass and stiffness. Increasing the **mass** of an object generally lowers its natural frequency, making it oscillate more slowly. Conversely, increasing the **stiffness** (or tension) of an object raises its natural frequency, causing it to oscillate faster. Engineers must calculate these frequencies carefully when designing structures like bridges or buildings to ensure that wind or traffic forces do not accidentally match their natural frequencies and cause resonance.

Complete the sentence: Resonance happens when the __________ matches the natural frequency, causing the amplitude of response to become very large.

Word bank: driving frequency · natural frequency · amplitude · stiffness · mass

Activity

Finding Resonance with a Pendulum
You will need: a string, a heavy bob (like a metal washer or small weight), a ruler, and a stopwatch.

- Tie the bob to one end of the string and secure the other end so it hangs freely. This is your pendulum.
- Measure the length of the string from the pivot point to the centre of the bob. Record this as L.
- Pull the bob aside slightly (less than 10 degrees) and release it without pushing. Count how many full swings (back and forth) occur in 30 seconds.
- Calculate the frequency of this free oscillation using the formula: frequency = number of swings / time in seconds. This is approximately the natural frequency for this length.
- Now, try to push the pendulum gently at regular intervals. Start by pushing it very slowly (low driving frequency).
- Gradually increase the speed of your pushes until you notice the swing going higher with each push. The point where the swing grows largest is where resonance has occurred.
- Record the length of the string and whether a longer or shorter string would have a higher natural frequency based on your observation.

String Length (m)	**Swings in 30s**	**Calculated Frequency (Hz)**	**Observation at Resonance**
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Questions

1. Define **natural frequency** in your own words.

2. Explain why a longer pendulum has a lower natural frequency than a shorter one.

3. Calculate the driving frequency required to cause resonance for an object with a natural frequency of 5 Hz. Show your working.

4. Why did the Tacoma Narrows Bridge collapse? Use the terms **driving frequency** and **amplitude** in your answer.

5. If you increase the stiffness of a guitar string, does its natural frequency go up or down? Explain why.

Extension

Research the Millennium Bridge in London. It experienced unexpected wobbling when it first opened because the walking pace of thousands of people matched the bridge's lateral natural frequency. How did engineers fix this problem without stopping the bridge from being used?